

Q17 /20

Choose the **two incorrect** options regarding SLOs (service level objectives) and SLAs (service level agreements) in the context of site reliability engineering.

- ☒ A. If an SLO is breached, there is a possibility of pursuing legal liability such as damages.
- ☒ B. Within the boundaries of meeting the SLO, the SRE team is allowed to intentionally cause outages if there are reasonable justifications.
- ☐ C. SLAs are usually set at values that are more difficult to achieve than SLOs.
- ☐ D. SLAs are established by considering factors such as the stability of competing services from other companies, the feasibility of service stability from an engineering perspective, and customer stability requirements.

Answer >

Q11

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered

Choose the correct description of database rollforward processing.

- A. ☐ Restore using a journal file before updating against a physical failure.
- B. ☐ Restore using a journal file after updating against a physical failure.
- C. ☐ Restore using a journal file before updating against a logical failure. ✓
- D. ☐ Restore using a journal file after updating against a logical failure.

Answer >

Questions

- Q1
- Q2
- Q3
- Q4
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered



Search



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13-1

OMEN

Q9 /20

4

/ 20

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Mr. Aoki is planning to visit the shrine for New Years. However, the parking lot at the shrine is very small. Once you park your car, you cannot leave until all of the cars that came after you leave. Choose the data structure that best represents the situation in this parking lot.

- A. ☐ stack ✓
- B. ☒ queue
- C. ☐ heap
- D. ☐ linked list

Answer >

Q4

Q5 Not Answered

Q6 Not Answered

Q7 Not Answered

Q8 Not Answered

Q9 Not Answered

Q10 Not Answered

Q11 Not Answered

Q12 Not Answered

Q13 Not Answered

Q14 Not Answered



Search

ENG
IN

Fill in the blanks of the following sentences with half-width digits.

I want to sort the sequence $a = [5, 2, 8, 3, 1, 6, 7, 9]$ in ascending order by exchanging adjacent numbers.
In this case, it is necessary to exchange at least times.

Answer >

- Q4
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered



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OMEN

You are teaching beginners A, B, C at your company. The beginners wrote the following functions. These functions receive an array of integers `a` and outputs the sum of all elements in array `a`.

Choose all functions that work properly. At least 1 function works properly.

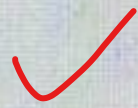
Beginner A

```
function sum(a) {  
  let answer = 0;  
  for (let i = 0; i < a.length; i++) {  
    answer += a[i];  
  }  
  return answer;  
}
```



Beginner B

```
function sum(a) {  
  let answer = 0;  
  for (let i = a.length; i > 0; i--) {  
    answer += a[i];  
  }  
}
```



Beginner C

```
function sum3(a) {  
  let answer = 0;  
  let i = 0;  
  while (i) {  
    if (i == a.length) {  
      break;  
    }  
    answer += a[i];  
    i++;  
  }  
  return answer;  
}
```

- A. ☐ Beginner A's function ✓
- B. ☐ Beginner B's function ✗
- C. ☐ Beginner C's function ✓


```
function sum(a) {  
  let answer = 0;  
  for (let i = 0; i < a.length; i++) {  
    answer += a[i];  
  }  
  return answer;  
}
```

Beginner B

```
function sum(a) {  
  let answer = 0;  
  for (let i = a.length; i > 0; i--) {  
    answer += a[i];  
  }  
  return answer;  
}
```



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/ 20

Completed

- A. ☐ The three `if` statements in lines 6-11 cause the program to be slow. We can speed up the program by replacing them with a single `for` statement and calling `rec` inside the statement.
- B. ☐ When `n` is equal to 1000, an integer overflow occurs and causes an infinite loop. If we use a real data type instead of a multiple precision integer type, we can prevent an infinite loop and then speed up the program.
- C. ☒ Although `rec(i)` always returns the same value, every time `rec(i)` is called, it calls the function three times recursively. That is why the program is so slow. We can speed up the program by storing the value `rec(i)` for each `i` when it is calculated for the first time, and returning that value every time it is called thereafter.
- D. ☐ None of the above.

Q11 Not Answered

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered

Answer >



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• 3, 1.

For this problem, you implemented a function `solve` as follows.
Note that all the variables are multiple precision integers.

```
function rec(i):  
    if i == 0:  
        return 1  
    else:  
        ans = 0;  
        if i >= 1:  
            ans += rec(i-1)  
        if i >= 2:  
            ans += rec(i-2)  
        if i >= 3:  
            ans += rec(i-3)  
        return ans  
  
function solve(n):  
    return rec(n)
```

Answer >

Q11 Not Answered

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered



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Completed

Choose the description that is **incorrect** regarding the differences between in-memory databases and traditional databases like MySQL or PostgreSQL.

- A. ☐ In-memory databases are often faster compared to relational databases like MySQL.
- B. ☐ In-memory databases are resilient to the physical failure of an HDD or SSD, making data loss less likely compared to storage-based databases. ✓
- C. ☐ In in-memory databases, ensuring sufficient main memory for storing all data is typically the only method used, rather than following rules like deleting infrequently used data to allocate enough capacity.
- D. ☐ Redis and Memcached are prominent examples of in-memory databases.

Answer >

Q11

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Q13

Q14 Not Answered

📍 Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered



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Choose the description that is **incorrect** regarding serverless architecture.

- A. ☐ Applications deployed in serverless architecture start only when needed, automatically scale down during idle periods, and can reduce costs.
- B. ☐ Billing in serverless architecture typically adjusts automatically based on factors like the number of requests, and in some cases, there may be no charges when there are no requests.
- C. ☒ Even when instances scale up, serverless architecture has the feature of zero-second initialization, allowing the application to start processing requests instantly.
- D. ☐ While serverless architecture allows you to be server-agnostic, fine tuning low-level settings below the OS layer on the servers where applications run is often more challenging than using IaaS.

Answer >

Q11

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

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Answer >

Q11

Q12

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Q16 Not Answered

📍 Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered

Microservice architecture provides many benefits over monolithic architecture. Which is/are the advantage(s) of microservice architecture? Select all correct options.

- ☒ A. Deployment in a microservice architecture is simpler because a microservice can be deployed for each service.
- ☒ B. Microservice architecture can reduce learning costs because it has a smaller codebase than monolithic systems.
- ☒ C. Microservice architecture can reduce single points of failure (SPOFs) because each service is loosely coupled.
- ☐ D. Microservice architecture can improve the total computational efficiency of the product because each service is minimal.

[Answer >](#)

Q11

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered



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Q19 /20

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/ 20

Complete

What are Docker images and Docker containers? Choose the correct option.

- A. ☐ Docker images are units of an OS, datafiles, and applications, including all dependencies between libraries used in the applications. A Docker container is an archive file that contains everything that can be used for Docker images.
- B. ☐ Docker containers are units of an OS, datafiles, and applications, including all dependencies between libraries used in the applications. A Docker image is an archive file that contains everything that can be used for Docker containers.
- C. ☐ Docker images are units of software that package code and all its dependencies. Docker containers are files used to create Docker images.
- ✓ ☒ Docker containers are units of software that package code and all its dependencies. Docker images are files used to create Docker containers.

Answer >

Q11

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered



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Which of the following is/are true about ChatOps? Choose all of the correct answers.

- ☒ A. It aims to create a centralized location where everyone's actions, tools, notifications, and diagnoses happen in full view.
- ☐ B. It primarily aims to expedite frequent communication, such as customer support, between information-technology operations (Ops) and end users using efficient chat tools.
- ☒ C. It contributes to creating a transparent team culture by bringing information that has been strongly dependent on a small set of employees to a centralized location.
- ☒ D. It is not only for technical teams but also for non-technical teams such as sales, marketing and finance. It can give a clear understanding of what is actually going on in the infrastructure and who is responsible for which systems and what each person does.

Answer >

Q11

Q12

Q13

Q14 Not Answered

Q15 Not Answered

Q16 Not Answered

Q17 Not Answered

Q18 Not Answered

Q19 Not Answered

Q20 Not Answered

neighboring array elements that have the same value in the array.

For example:

Given `arr = {0, 3, 2, 2, 6, -1, 3}`, the elements `2, 2` are next to each other and are the same value, so `true` is returned.

Given `arr = {0, 3, 2, 5, 6, -1, 3}`, there are no elements with the same values next to each other, so `false` is returned.

```
static boolean containConsecutive(int[] arr) {  
    for (int i = 0; i < [ ① ]; i++) {  
        if ([ ② ]) {  
            return true;  
        }  
    }  
    return false;  
}
```

Choose the appropriate formula to enter into [①], [②] to make the function work.

Answer >



Search



```
for (int i = 0; i < [ ① ]; i++) {  
    if ([ ② ]) {  
        return true;  
    }  
}  
return false;  
}
```

Choose the appropriate formula to enter into [①], [②] to make the function work.

- A. ☐ [①] `arr.length-1`, [②] `arr[i] == arr[i-1]`
- B. ☒ [①] `arr.length-1`, [②] `arr[i] == arr[i+1]` ✓
- C. ☐ [①] `arr.length`, [②] `arr[i] == arr[i-1]`
- D. ☐ [①] `arr.length`, [②] `arr[i] == arr[i+1]`
- E. ☐ All the above are incorrect

Answer >



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Age confirmation of participants

Read me when using external libraries

Memory Limit(MB) 512, Execution Timeout(ms) 2000

You will participate in a meetup with your fellow hobbyists. You are in charge of organizing the after-party for this meetup. In your country, the age considered to be of legal adulthood is (x) , and anyone under the age of (x) is considered a minor and is not allowed to drink any beverages containing alcohol.

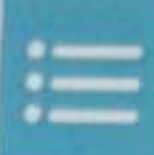
For the after-party, the group will order an all-you-can-drink course. If there are any underage participants at the after-party, everyone will order the all-you-can-drink course that includes only soft drinks. On the other hand, if there are no underage participants, everyone will order the all-you-can-drink course that includes alcoholic beverages.

Test Out

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Age confirmation of participants



the age of (x) is considered a minor and is not allowed to drink any beverages containing alcohol.

For the after-party, the group will order an all-you-can-drink course. If there are any underage participants at the after-party, everyone will order the all-you-can-drink course that includes only soft drinks. On the other hand, if there are no underage participants, everyone will order the all-you-can-drink course that includes alcoholic beverages.

Since it would be a hassle to go through the age verification process on-site, you decided to check the list of ages of the participants in advance to figure out the all-you-can-drink course you should choose.

Given the number of participants n , the age at which a participant will be

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13 if

Test Output



Search

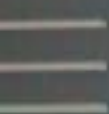
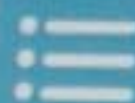
Age confirmation of participants

participants, everyone will order the all-you-can-drink course that includes alcoholic beverages.

Since it would be a hassle to go through the age verification process on-site, you decided to check the list of ages of the participants in advance to figure out the all-you-can-drink course you should choose.

Given the number of participants n , the age at which a participant will be considered to be of legal age x , and a list of participants' ages $y_1, \dots, y_n$, create a program to output whether the all-you-can-drink course will include alcohol or be limited to only soft drinks.

Implementation Details



- 4
- 3
- 4
- 5
- 6
- 7
- 8
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- 10
- 11
- 12
- 13

Test Output



Search

Age confirmation of participants

Given participants ages $y_1, \dots, y_n$, create a program to output whether the all-you-can-drink course will include alcohol or be limited to only soft drinks.

Implementation Details

CLI

Implement a command line application that receives standard input and writes the answer to its standard output. For details, see the "Command line application template" section at the bottom of the page.

Input Rules

The program is executed as follows:

```
myApp < input.in
```



Search

page.

Input Rules

The program is executed as follows:

```
./myApp < input.in
```

The standard input format is as follows:

```
 $n$   $x$   
 $y_1$   $y_2$   $\dots$   $y_n$ 
```

In the first line, a space-separated list of the number of participants n , and the age at which they are considered to be of legal age x , are given.

In the second line, a space-separated list of the ages of the participants is given, $(1 \leq i \leq n)$.

page.

Input Rules

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Age confirmation of participants

```
n x
y1 y2 ... yn
```

In the first line, a space-separated list of the number of participants n , and the age at which they are considered to be of legal age x , are given.

In the second line, a space-separated list of the ages of the participants is given, $y_i (1 \leq i \leq n)$.

Restrictions:

- $1 \leq n \leq 100$
- $10 \leq x \leq 21$
- $10 \leq y_i \leq 120 (1 \leq i \leq n)$
- All inputs are integers.

Output Rules

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Age confirmation of participants

```
n x
y1 y2 ... yn
```

In the first line, a space-separated list of the number of participants n , and the age at which they are considered to be of legal age x , are given.

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Output Rules

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Age confirmation of participants

- $1 \leq n \leq 100$
- $10 \leq x \leq 21$
- $10 \leq y_i \leq 120 (1 \leq i \leq n)$
- All inputs are integers.

Output Rules

Your CLI program should write the answer to its standard output.

The format of the standard output is as below:

- If there are any values in the list of ages that are less than the age considered to be of legal adulthood, output "Soft Drink".
- If there are no values under the age considered to be of legal adulthood in the list of ages, output "Alcohol".

Examples

Test Output

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Age confirmation of participants

- $1 \leq n \leq 100$
- $10 \leq x \leq 21$
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Output Rules

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Examples

Test Output

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Command line application template for Python3.x

Implement CLI application by editing `main.py`.

You may add new files to keep your code clean, if it is allowed in your challenge.

How to get stdin lines

In `main.py`, there is a function called `main`, which gives stdin lines as `lines`.

```
def main(lines):  
    # code to run
```



Search

Command line application template for Python3.x

Implement CLI application by editing `main.py`.

You may add new files to keep your code clean, if it is allowed in your challenge.

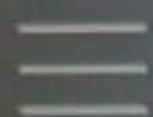
How to get stdin lines

In `main.py`, there is a function called `main`, which gives stdin lines as `lines`.

```
def main(lines):  
    # code to run
```



Search



Age confirmation of participants



under 20 years old.

Example 2

Standard input (`00_sample_02.in`)

```
3 18
120 18 65
```

Standard output

```
Alcohol
```

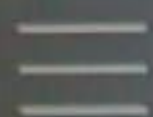
Some expected inputs and outputs are defined in `test/basic_testcases.json`. Please refer to them when implementing.



Command line



Search



Age confirmation of participants



under 20 years old.

Example 2

Standard input (`00_sample_02.in`)

```
3 18
120 18 65
```

Standard output

```
Alcohol
```

Some expected inputs and outputs are defined in `test/basic_testcases.json`. Please refer to them when implementing.



Command line



Search

Age confirmation of participants

the list of ages, output "Alcohol".

I/O Examples

Example 1

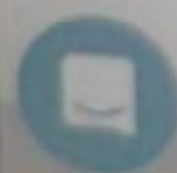
Standard input (`00_sample_01.in`)

```
5 20
25 23 28 19 20
```

Standard output

```
Soft Drink
```

Output "Soft Drink" because the list of ages includes participants whose age is under 20 years old.



Feedback



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Age confirmation of participants

the list of ages, output "Alcohol".

I/O Examples

Example 1

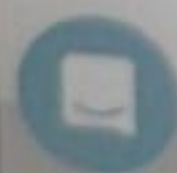
Standard input (`00_sample_01.in`)

```
5 20
25 23 28 19 20
```

Standard output

```
Soft Drink
```

Output "Soft Drink" because the list of ages includes participants whose age is under 20 years old.



Feedback

Test Output



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